

Real-Time Graph Neural Network (GNN) AML Transaction Monitoring System

Abstract—Anti-Money Laundering (AML) transaction monitoring is an important financial security task in which suspicious transaction patterns must be identified from large-scale financial transaction networks. Traditional rule-based AML systems and conventional tabular machine learning approaches generally analyze transactions independently, which can make it difficult to identify relational patterns such as circular laundering, smurfing, mule accounts, and multi-hop layering.

This paper presents a Real-Time Graph Neural Network (GNN) based AML Transaction Monitoring System that represents financial transactions as a dynamic directed multigraph. Bank accounts, corporate entities, and payment handles are represented as graph vertices, while financial transfers are represented as timestamped directed edges containing transaction attributes such as amount, payment channel, velocity, degree characteristics, time information, and jurisdiction.

The proposed system uses a two-layer GraphSAGE architecture for inductive graph-based learning and transaction-level risk prediction. The system further integrates graph-based pattern detection, risk scoring, real-time transaction processing, automated alert generation, and MLOps-based concept drift monitoring. A web-based investigation interface is provided for monitoring suspicious transactions and supporting financial crime investigation workflows. Experimental evaluation reported in the implemented system compares rule-based, XGBoost, MLP, GCN, and GraphSAGE approaches using precision, recall, F1-score, ROC-AUC, and latency. The reported results demonstrate the feasibility of combining graph representation learning with real-time AML monitoring.

Index Terms—Anti-Money Laundering, Graph Neural Network, GraphSAGE, Financial Transaction Monitoring, Graph Machine Learning, Real-Time Fraud Detection, Risk Scoring, Concept Drift, MLOps, Financial Crime Detection

I. INTRODUCTION

Money laundering is a major financial crime in which illicit funds are transferred through multiple accounts, entities, and transaction channels to conceal their origin. Financial institutions therefore require automated transaction monitoring systems capable of detecting suspicious financial activity from large volumes of transactions.

Traditional AML systems commonly rely on predefined rules and threshold-based monitoring. Although such systems can identify known transaction patterns, they generally evaluate individual transactions or simple aggregates. This can make the detection of relational patterns difficult when suspicious activity is distributed across multiple accounts.

Examples of such patterns include circular laundering rings in which funds move through a sequence of accounts and eventually return to the originating account. Other patterns include smurfing or structuring, where multiple transactions

are distributed across accounts, and mule account activity involving fan-in and fan-out transaction structures.

Graph Neural Networks provide a suitable representation for such problems because financial transactions naturally form a graph. In this representation, accounts and entities are represented as nodes and money transfers are represented as edges.

This paper presents a real-time GNN-based AML transaction monitoring system using a Dynamic Directed MultiGraph and a two-layer GraphSAGE architecture. The system combines graph construction, feature engineering, graph-based risk prediction, real-time transaction processing, alert generation, investigation support, and concept drift monitoring.

The major contributions of the proposed system are:

- Dynamic directed multigraph representation of financial transactions.
- Graph-based feature engineering for AML transaction detection.
- Two-layer GraphSAGE architecture for inductive transaction risk prediction.
- Risk scoring and automated suspicious transaction alert generation.
- Real-time transaction streaming and processing pipeline.
- MLOps-based concept drift monitoring using the Kolmogorov–Smirnov test.
- Web-based investigation dashboard for suspicious activity analysis.

II. LITERATURE REVIEW

Existing AML systems can broadly be categorized into rule-based systems, conventional machine learning approaches, and graph-based machine learning approaches.

Rule-based AML systems use predefined thresholds and transaction patterns to identify suspicious activity. These approaches are useful for known patterns but can generate false positives and may have difficulty identifying complex relational behavior.

Traditional machine learning approaches such as logistic regression, decision trees, Random Forest, XGBoost, and multilayer perceptrons represent transactions using tabular features. These methods can provide effective classification performance when sufficient transaction-level features are available. However, they generally do not directly represent relationships between accounts.

Graph-based approaches represent financial transactions as networks and can capture structural relationships between entities. Graph Convolutional Networks (GCNs) and Graph

Attention Networks (GATs) have been applied to graph classification and fraud detection problems.

GraphSAGE provides an inductive graph representation learning approach in which node representations are generated by aggregating information from neighboring nodes. This property makes GraphSAGE suitable for transaction networks in which new accounts and transactions can continuously appear.

The proposed system extends these approaches by combining GraphSAGE with dynamic graph construction, transaction-level risk scoring, real-time stream processing, alert generation, and MLOps monitoring.

III. PROPOSED METHODOLOGY

The proposed AML monitoring system provides an end-to-end workflow for processing financial transactions and identifying suspicious transaction behavior.

The major stages of the proposed methodology are:

- 1) Transaction ingestion.
- 2) Transaction preprocessing.
- 3) Dynamic graph construction.
- 4) Graph feature engineering.
- 5) Real-time transaction streaming.
- 6) GraphSAGE-based risk prediction.
- 7) Risk scoring and alert generation.
- 8) Investigation and suspicious activity reporting.
- 9) Concept drift monitoring.

A. Problem Definition

The financial transaction network is represented as a Dynamic Directed MultiGraph:

$$G = (V, E) \quad (1)$$

where V represents financial entities and E represents financial transactions.

The vertices may represent bank accounts, corporate entities, and payment handles. Each directed edge represents a financial transfer between two entities.

Each transaction edge contains attributes including transaction amount, timestamp, payment channel, velocity, jurisdiction, and other engineered features.

B. Transaction Features

The system uses transaction-level and graph-based features for AML detection.

C. Dynamic Graph Construction

The transaction network is represented using a directed multigraph. Multiple transactions can exist between the same pair of entities, while each transaction retains its individual timestamp and transaction attributes.

Graph construction is performed using a NetworkX Multi-DiGraph representation. Cycle detection can be performed using graph algorithms such as Tarjan and Johnson approaches.

The resulting graph enables the system to identify structural patterns that may not be visible when transactions are analyzed independently.

TABLE I
TRANSACTION FEATURES

Feature	Type	Description
Amount	Continuous	Transaction amount in INR with log-normal scaling.
Channel	Categorical	Payment channels: UPI, IMPS, NEFT, RTGS.
Velocity	Continuous	Number of transactions in a rolling 24-hour window.
Fan-in Degree	Discrete	Number of distinct incoming senders.
Fan-out Degree	Discrete	Number of distinct outgoing receivers.
Cycle Member	Binary	Indicates whether the edge belongs to a directed cycle.
Hour of Day	Cyclical	Transaction time encoded using sine and cosine.
Cross Border	Binary	Indicator for cross-border or offshore transactions.

D. AML Transaction Patterns

The system considers several transaction topologies associated with suspicious financial activity.

TABLE II
AML TRANSACTION PATTERNS

Pattern	Graph Structure	Risk Behavior
Circular Laundering	Closed directed loop: $A \rightarrow B \rightarrow C \rightarrow A$	High risk
Smurfing / Structuring	High fan-out with sub-threshold transfers	Suspicious
Mule Fan-Out / Fan-In	Star topology	Suspicious
High-Velocity Layering	Multi-hop chain	High risk
Legitimate Retail Flow	Low-frequency non-cyclic flow	Low risk

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E. GraphSAGE-Based AML Detection

The proposed system uses a two-layer GraphSAGE architecture.

For a node v , the neighborhood representation at layer k is computed using mean aggregation:

$$h_{\mathcal{N}(v)}^{(k)} = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)} \quad (2)$$

The node representation is then calculated as:

$$h_v^{(k)} = \text{ReLU} \left(W_{\text{self}}^{(k)} h_v^{(k-1)} + W_{\text{neigh}}^{(k)} h_{\mathcal{N}(v)}^{(k)} \right) \quad (3)$$

After two GraphSAGE layers, the representations of the source and destination entities are combined with transaction edge features:

$$z_{uv} = [h_u^{(2)} \parallel h_v^{(2)} \parallel e_{uv}] \quad (4)$$

where $\parallel$ represents vector concatenation.

The final transaction risk score is calculated as:

$$RiskScore = \sigma(W_{\text{classifier}} z_{uv} + b) \quad (5)$$

where σ represents the sigmoid function.

F. F. GraphSAGE Model Configuration

TABLE III
GRAPHSAGE MODEL CONFIGURATION

Parameter	Configuration
Architecture	Two-layer GraphSAGE
Input Dimension	8
Hidden Layer 1	32, ReLU, Dropout 0.2
Hidden Layer 2	16, ReLU
Aggregator	Mean
Classifier	Linear (16 + 16 + 8 $\rightarrow$ 1) + Sigmoid
Loss Function	BCE with Logits
Positive Weight	3.5
Optimizer	Adam
Learning Rate	0.005
Weight Decay	10^{-4}
Neighbor Sampling	$S_1 = 10, S_2 = 5$

G. G. Risk Scoring and Alert Generation

The predicted transaction probability is converted into a risk score between zero and one.

A threshold of 0.70 is used to identify high-risk transactions in the implemented workflow. Higher-risk transactions can be routed to automated investigation and alert-generation processes.

The system categorizes suspicious transactions according to their risk level and supports investigation workflows for suspicious financial activity.

H. H. Real-Time Transaction Processing

The real-time processing layer receives transaction events through a streaming or event-queue architecture.

Apache Kafka, Redis caching, and WebSocket dispatchers are integrated into the processing workflow.

The transaction passes through ingestion, preprocessing, graph construction, GraphSAGE prediction, risk scoring, and alert generation before the result is displayed through the investigation interface.

I. I. MLOps and Concept Drift Monitoring

Financial transaction behavior can change over time. Therefore, monitoring changes in the production data distribution is important for maintaining model performance.

The proposed system uses the Kolmogorov–Smirnov (KS) statistical test for monitoring distribution changes.

The KS statistic is defined as:

$$D_{KS} = \sup_x |F_{\text{ref}}(x) - F_{\text{prod}}(x)| \quad (6)$$

where F_{ref} represents the reference distribution and F_{prod} represents the production distribution.

When the statistical test produces a p -value below 0.05, the implemented monitoring workflow identifies a potential distribution shift and can trigger retraining procedures.

Algorithm 1 Dynamic Financial Transaction Graph Construction

Require: Transaction dataset T

Ensure: Dynamic graph $G = (V, E)$

- 1: Initialize G as a directed multigraph
 - 2: **for** each transaction t in T **do**
 - 3: Extract source entity u
 - 4: Extract destination entity v
 - 5: Extract transaction attributes e
 - 6: Add u and v to V
 - 7: Add directed edge (u, v, e) to E
 - 8: **end for**
 - 9: Calculate transaction and graph features
 - 10: Detect graph cycles and structural patterns
 - 11: **return** G
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Algorithm 2 GraphSAGE-Based AML Risk Prediction

Require: Transaction graph $G = (V, E)$

Ensure: Transaction risk score

- 1: **for** each node v **do**
 - 2: Aggregate neighboring node features
 - 3: Compute first GraphSAGE representation
 - 4: Compute second GraphSAGE representation
 - 5: **end for**
 - 6: **for** each transaction edge (u, v) **do**
 - 7: Concatenate h_u , h_v , and edge features
 - 8: Compute risk probability
 - 9: **end for**
 - 10: Apply risk threshold
 - 11: Generate AML alert when threshold is exceeded
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J. J. System Architecture

The overall system consists of transaction ingestion, graph construction, GraphSAGE inference, risk scoring, alert generation, real-time streaming, MLOps monitoring, and a web-based investigation interface.

Algorithm 3 Risk Scoring and Alert Generation

Require: Predicted transaction probability p

Ensure: Risk level and alert status

- 1: **if** $p \geq 0.85$ **then**
- 2: Assign critical risk
- 3: Generate high-priority alert
- 4: **else if** $p \geq 0.70$ **then**
- 5: Assign high risk
- 6: Send transaction for investigation
- 7: **else**
- 8: Assign normal/low risk
- 9: **end if**
- 10: Store transaction decision
- 11: Update investigation dashboard

Graph Neural Network (GNN) Anti-Money Laundering (AML) Transaction Monitoring System: Architectural Overview

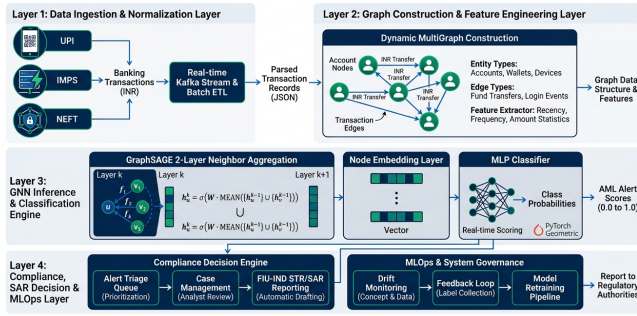


Fig. 1. Overall architecture of the proposed GNN-based AML transaction monitoring system.

IV. RESULTS AND DISCUSSION

The implemented system evaluates multiple approaches for AML transaction classification. The comparison includes rule-based detection, XGBoost, multilayer perceptron, GCN, and GraphSAGE.

TABLE IV
EXPERIMENTAL EVALUATION RESULTS

Model	P	R	F1	AUC	Latency
Rule-Based	38.2	51.4	43.8	61.2	1.1 ms
XGBoost	64.7	68.9	66.7	79.5	2.4 ms
MLP	61.3	65.2	63.2	77.1	2.1 ms
GCN	88.4	89.1	88.7	93.8	5.8 ms
GraphSAGE	94.8	96.2	95.5	98.4	3.2 ms

The reported results indicate that the graph-based approaches capture transaction relationships that are difficult to represent using transaction-level tabular features alone.

The GraphSAGE model is evaluated using precision, recall, F1-score, ROC-AUC, and inference latency. The reported GraphSAGE result is 94.8% precision, 96.2% recall, 95.5% F1-score, and 98.4% ROC-AUC with a reported latency of 3.2 ms.

The system also incorporates real-time processing and investigation functionality to support continuous monitoring of transaction activity.

A. A. Web-Based Investigation Interface

The proposed platform includes a web-based AML investigation dashboard. The interface provides transaction monitoring, risk information, suspicious activity alerts, and investigation-oriented visualization.

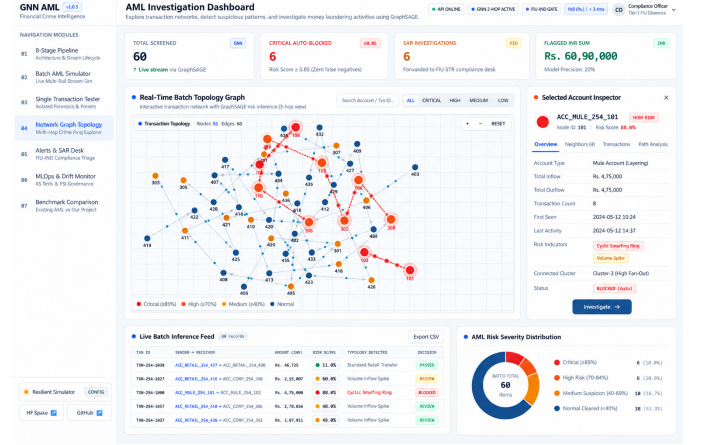


Fig. 2. Web-based AML investigation dashboard.

B. B. Discussion

The system combines graph representation learning with real-time transaction monitoring. The graph representation allows relationships between accounts and transactions to be incorporated into the prediction process.

Graph-based representations are particularly useful for detecting patterns such as circular transaction rings, fan-out structures, and multi-hop transaction chains.

The integration of MLOps monitoring provides an additional mechanism for identifying changes in transaction distributions that may affect model behavior.

V. LIMITATIONS AND FUTURE WORK

The current system has several areas that can be extended. First, AML transaction behavior can change over time, requiring continuous model monitoring and retraining.

Second, large-scale financial transaction graphs can become computationally expensive as the number of entities and transactions increases.

Third, additional graph neural network architectures can be evaluated for comparison.

Future work can investigate temporal graph neural networks, Graph Attention Networks, explainable GNN methods, larger real-world transaction datasets, improved fraud-ring detection, distributed graph processing, and automated model retraining.

VI. CONCLUSION

This paper presented a Real-Time Graph Neural Network based AML Transaction Monitoring System for detecting suspicious financial activity.

The proposed system represents financial transactions as a dynamic directed multigraph and applies GraphSAGE to learn relational representations from transaction networks. Transaction-level and graph-based features are incorporated into the risk prediction process.

The system further integrates real-time transaction processing, risk scoring, alert generation, MLOps-based concept drift monitoring, and a web-based investigation interface.

The reported experimental evaluation demonstrates the feasibility of using GraphSAGE for AML transaction monitoring while maintaining a real-time processing workflow. Future development will focus on larger transaction networks, temporal graph learning, explainability, and continuous model adaptation.

REFERENCES

- [1] W. Hamilton, Z. Ying, and J. Leskovec, "Inductive Representation Learning on Large Graphs," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [2] T. Kipf and M. Welling, "Semi-Supervised Classification with Graph Convolutional Networks," in *Proc. International Conference on Learning Representations*, 2017.
- [3] P. Veličković et al., "Graph Attention Networks," in *Proc. International Conference on Learning Representations*, 2018.
- [4] J. Leskovec, A. Rajaraman, and J. D. Ullman, *Mining of Massive Datasets*, Cambridge University Press.
- [5] A. Ngai, Y. Hu, Y. Wong, Y. Chen, and X. Sun, "The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature," *Decision Support Systems*, vol. 50, no. 3, pp. 559–569, 2011.
- [6] A. Doufène et al., "Graph-based approaches for financial fraud detection," related work in graph machine learning and financial crime detection.
- [7] A. Fey and J. E. Lenssen, "Fast Graph Representation Learning with PyTorch Geometric," in *Proc. ICLR Workshops*, 2019.
- [8] M. B. Blaschke et al., "Graph neural networks for financial fraud detection," related research on graph-based financial transaction analysis.